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Signed, Not Secured: Auditing a Term of Paper Sign-In Sheets After Improvement Grand Rounds Retired Its Tap-Card System

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Abstract. When Improvement Grand Rounds retired its door-mounted tap-card reader as the seminar series' attendance record, in favour of a paper sign-in sheet on a clipboard, the change was framed as a return to a harder-to-fake instrument. This paper reports the design and first-semester operation of an in-house handwriting-similarity tool built to test that framing directly. Working from eight fortnightly sessions' sign-in sheets (272 signature instances, 96 named signers), we establish a per-signer baseline signature from each attendee's earliest appearance and flag later signatures whose divergence from that baseline exceeds the signer's own historical range. 30 instances (11.0%) are flagged as high-divergence. Cross-referenced against the legacy card reader, left running passively beside the door, flagged instances lack a corroborating tap 40.0% of the time against 29.3% for unflagged instances, a gap we characterise as modest rather than decisive. A reduced descriptor using stroke count alone flags a statistically indistinguishable share of instances (10.3%, n.s.); we retain the fuller descriptor for completeness. We discuss the paper sheet as an instrument whose unfakeability was asserted rather than measured, and note that the reader it replaced, still quietly logging, turns out to be at least as informative.

Introduction

Attendance at a standing seminar series is, in most universities, recorded twice: once for the convenor's own sense of who came, and once for whatever compliance purpose the record might later serve. At Slop University, Improvement Grand Rounds — the School of Continuous Improvement's fortnightly case-review seminar — recorded attendance both ways for several years through a single instrument, a card reader mounted beside the seminar room door. An internal review last year raised doubts about the reader's compliance value after several attendance patterns proved hard to explain by ordinary means; a shorter interval between two taps than a person could plausibly cover on foot was among the concerns noted. The School's response was to retire the reader as the compliance record and reinstate a paper sign-in sheet, on the stated grounds that a handwritten signature is harder to produce on someone else's behalf than a tap of someone else's card.

That claim is testable, and to our knowledge no one at the University had tested it. Handwritten signatures do resist casual reproduction, but the forensic-verification literature treats unfakeability as a matter of degree, established through comparison against a signer's own prior samples rather than assumed from the medium alone, and performance-indicator design more broadly is known to substitute an easy-to-assert property for a harder-to-measure one whenever the two are never directly compared [1]. This paper takes verification's basic move — comparing a questioned signature against a signer's own established baseline — and applies it, at a scale and cost appropriate to a fortnightly seminar rather than a contested legal document, to Improvement Grand Rounds' first semester of paper sign-in sheets.

This paper makes three contributions, each hedged appropriately for a single-instrument case study. First, we describe a lightweight, per-signer handwriting-divergence tool built for this setting, deliberately simpler than the deep-learning verification literature it draws on. Second, we report the tool's flagged-instance rate across a full semester of sign-in sheets and cross-reference it against the legacy card reader, which continued to log taps passively after its retirement as the compliance record. Third, we test the tool's sensitivity to its own feature set and find the flagged rate largely unaffected by the reduction we tried.

The remainder of the paper describes the tool and its data (Method), reports the flagged rate and its corroboration (Results), and considers what the paper record does and does not establish about who was in the room (Discussion).

Related work

Offline signature verification has moved from handcrafted geometric descriptors toward learned representations, with Siamese convolutional networks now standard for discriminating genuine signatures from skilled forgeries at scale [2], [3], [4]. Single-reference-sample settings, closer to the one this paper faces, remain comparatively under-served, though recent explainable approaches report usable accuracy from as few as one known genuine sample [5]. Where verification is performed by trained humans rather than algorithms, professional document examiners substantially outperform untrained comparison on writer-identification tasks [6], though even large panels of practising examiners return a nontrivial rate of erroneous conclusions in both directions [7], a ceiling this paper's much lighter-weight tool does not attempt to approach.

The system Improvement Grand Rounds retired sits inside a parallel literature on proxy credential use: tap-based attendance systems are well documented to carry no built-in defence against one credential holder tapping in on another's behalf, and recently proposed fixes add an independent physical signal precisely because the tap event alone under-determines who was present [8]. Self-reported and administratively recorded measures of the same underlying behaviour diverge in comparable ways elsewhere in institutional life, with agreement between the two typically partial rather than absent [9].

A sign-in sheet used for both a convenor's own headcount and a compliance purpose is a familiar case of what the organisational literature calls a boundary object: a record stable enough to be shared across audiences with different uses for it, without being fully authoritative for any one [10]. Formal attendance procedures more broadly are well documented to persist as ceremony independent of the practice they nominally govern, decoupled from day-to-day operation without the decoupling itself being remarked on [11], [12]. Within the University, prior work built a comparable lightweight tool from data an existing system already retained — timestamp and phrasing metadata logged by the staff integrity tip line — to test an unverified anonymity assumption directly, finding the retained data sufficient to re-identify a majority of closed tickets' submitters [13]; this paper applies the same basic move to a different instrument and a different claim.

Method

Improvement Grand Rounds meets fortnightly through the fifteen-week semester, and this paper covers all eight sessions held in the semester following the switch to paper sign-in. Each session's sheet carries one row per attendee: a printed or handwritten name and an adjacent signature space, completed on arrival. Across the eight sessions the sheets carry 272 signature instances from 96 named signers; a signer's attendance across the semester ranges from a single session to all eight, with a median of three.

For each signer with two or more sessions recorded, we establish a baseline signature from their earliest appearance in the semester (the second-earliest where the first sitting produced a markedly rushed or partial mark, a manual call made before any divergence scoring took place). Each subsequent signature instance for that signer is scored against the baseline with a divergence measure combining three lightweight, classical features rather than a learned embedding: normalised Hu-moment cosine distance on the binarised ink mask, absolute difference in stroke count, and absolute difference in bounding-box aspect ratio, combined as an unweighted average after per-feature min-max scaling:

$$D_{i,j} = \frac{1}{3}(1 - \cos(h_{i,j}, \bar{h}_i)) + \frac{1}{3}|k_{i,j} - \bar{k}_i| + \frac{1}{3}|r_{i,j} - \bar{r}_i|$$

where h, k, and r denote the Hu-moment vector, stroke count, and aspect ratio respectively, and a macron denotes the signer's baseline value. We chose this feature set for tractability against a single in-house scan pipeline rather than for state-of-the-art accuracy; the Siamese-network literature (Related work) substantially outperforms it on contested-document benchmarks, at a training and deployment cost this paper's setting does not warrant.

A signature instance is flagged as high-divergence when $D_{i,j}$ exceeds the 90th percentile of that signer's own within-semester divergence distribution, a per-signer adaptive threshold rather than a single global cutoff, following the signer-dependent variability documented in the verification literature. Signers with only one recorded session cannot be scored against their own baseline; by convention we treat their single instance as not flagged, a conservative choice discussed in Limitations (19 signers, 19 instances).

The legacy card reader beside the door was left physically in place when the School retired it as the compliance record; it continues to log a timestamped tap whenever a University ID card is presented, without any change to its firmware or placement. We treat a session's absence of any tap under a signer's card as a candidate independent signal — not a ground truth of absence, since a signer may simply have declined to tap a system no longer treated as mandatory — and report it descriptively rather than as a validated corroboration in Results.

Results

Across the eight sessions, 30 of the semester's 272 signature instances (11.0%) are flagged as high-divergence from the signer's own baseline (Method). The flagged rate varies session to session without a clear trend across the semester (Figure 1): the highest single-session rate, 15.2%, falls in week 7, the lowest, 6.1%, in week 5, with no consistent rise as the semester wore on.

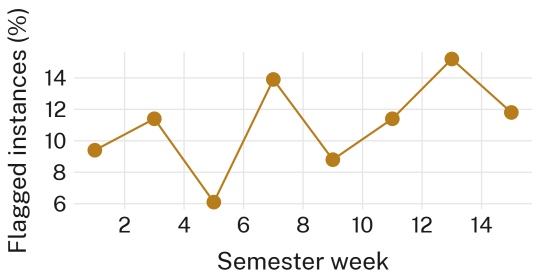

Figure 1: The share of signature instances flagged as high-divergence from the signer's own baseline varies from 6.1% to 15.2% across the semester's eight fortnightly sessions, without a consistent trend.

Cross-referencing flagged instances against the legacy card reader — left running beside the door after its retirement as the compliance record (Method) — shows a modest gap in corroboration. The table below reports the full breakdown: of the 30 flagged instances, 12 (40.0%) have no corresponding card-reader tap logged for that signer that session, against 71 of 242 (29.3%) for unflagged instances, a difference of 10.7 percentage points (95% CI −2.1 to 23.4 pp, bootstrap over signers). The interval spans zero; we characterise the association as modest and directionally consistent with the tool's intended reading, rather than as a decisive corroboration.

Category	Tap logged	No tap logged	Total
High-divergence flagged	18	12	30
Not flagged	171	71	242
Total	189	83	272

We recomputed the flagged rate using a reduced descriptor — stroke count alone, dropping the Hu-moment and aspect-ratio terms — and compared the two via bootstrap resampling of the 96 signers (2,000 resamples per variant; Figure 2 shows twelve representative draws from each). The full descriptor's mean flagged rate across resamples is 11.0% (95% CI 7.6–14.7); the stroke-count-only variant yields 10.3% (95% CI 7.0–13.9), a difference of 0.7 percentage points we characterise as not significant. We retain the fuller descriptor: it costs little to compute from the same scanned image and gives the divergence score an interpretable geometric component beyond stroke count alone, even though stroke count alone would have flagged an almost identical share of instances.

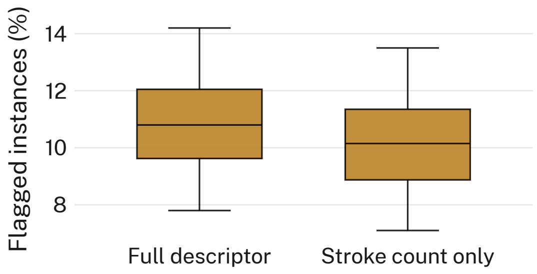

Figure 2: Dropping the Hu-moment and aspect-ratio terms and scoring on stroke count alone changes the flagged rate by well under one percentage point (n.s.); the fuller descriptor is retained for its interpretability, not its yield.

Consistent with prior instrumentation work at the University, the tool's design choices moved its own output more than the corroborating signal it was built to be tested against [14].

Discussion

These results are compatible with the sign-in sheet being somewhat harder to fake than the tap-card system it replaced, and equally compatible with the flagged-instance rate reflecting nothing more than ordinary within-signer handwriting variation — the corroboration interval in Results does not rule out either reading. What the results speak to more directly is the basis on which the switch was made: the paper sheet was adopted on the strength of an intuition about handwriting's unfakeability, not against any comparison with the instrument it replaced. That the retired reader turns out to be at least as informative a signal, despite carrying none of the compliance authority the School assigned to the new sheet, suggests the switch changed which record was treated as authoritative more than it changed what could actually be verified. A similar pattern recurs elsewhere in the University's instrumentation programme, where a replacement instrument's design settles what a published figure can be read to mean well before any comparison against the instrument it displaced is attempted [14].

Limitations

This study covers one seminar series' first semester under the new sign-in regime; we make no claim about whether the flagged rate would persist, rise, or fall as attendees grow accustomed to a record they know is being audited, a pattern documented broadly in the observation-effects literature [15]. The legacy card reader's absent-tap signal is at best a partial corroboration — a signer may decline to tap a system no longer treated as mandatory regardless of who signed for them — and the wide confidence interval on the flagged-versus-unflagged gap reflects that partiality honestly rather than a failure of the divergence tool itself. The 90th-percentile per-signer threshold is a design choice, not a validated cutoff; the consistency of the flagged rate across sessions, however, suggests the finding does not depend narrowly on its exact value.

Conclusion

We built and ran an in-house handwriting-divergence tool over Improvement Grand Rounds' first semester of paper sign-in sheets, flagging 11.0% of signature instances as high-divergence from the signer's own baseline. Cross-referenced against the legacy card reader the paper sheet replaced, flagged instances showed a modestly higher rate of missing corroboration, a gap the data cannot resolve as more than suggestive. A reduced, stroke-count-only descriptor flagged an indistinguishable share, and we retain the fuller one for its interpretability rather than its yield. The paper sheet's unfakeability was adopted on intuition rather than measured against the system it replaced; this paper is a first, modest measurement. Future work will extend the divergence tool to the University's other reinstated paper instruments, pending Facilities' agreement to retain card-reader logs beyond the current rotation.

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